SOMAIYA VIDYAVIHAR UNIVERSITY

16-05-2025 (M)

SET-E

Semester: January 2025– April 2025			
Maximum Marks: 50 **Examination: End-Semester Examination** **Duration: 2 Hrs.**			
Programme code: H05 **Programme:** Honors in Software and Mobile Application Development	**Class:** SY		**Semester:** IV (SYU-2023)
Institute/School/Department: K. J. Somaiya School of Engineering	**Name of the department/Section/Center:** COMP		
Course Code: 216H05C401	**Name of the Course:** Fundamental of Swift		
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question	Max. Marks
Q.1	Attempt any **four**	20
i)	Define a protocol with an associated type and implement it in a struct.	05
ii)	What is the purpose of the **`throws`** keyword in Swift functions? Provide a throwing function.	05
iii)	What is a class-only protocol? Show with the help of an how to define and use one.	05
iv)	How is `as?` different from `as!` in type casting? Illustrate with a code snippet.	05
v)	Write a Swift struct that conforms to the **`Hashable`** protocol and explain its significance.	05
vi)	Explain the role of generics in enhancing type safety with an example.	
Q.2	Attempt any **one**	10
i)	Implement a simple calculator that uses closures for different operations. – Create an **`enum`** named **`Operation`** with cases for **`add`**, **`subtract`**, **`multiply`**, and **`divide`**. – Write a function named **`calculate`** that takes two **`Double`** values and an **`Operation`** enum case as input. – Inside the **`calculate`** function, define a dictionary where the keys are the **`Operation`** enum cases and the values are closures that perform the corresponding arithmetic operation on two **`Double`** inputs and return a **`Double`**. – Use a **`switch`** statement on the **`Operation`** to retrieve the appropriate closure from the dictionary and execute it with the provided numbers. – Handle the division by zero case by returning **`nil`** (make the function return an optional **`Double`**). Demonstrate calling the **`calculate`** function with different operations and numbers, including a division by zero scenario. Use trailing closure syntax when calling the closures indirectly through the dictionary.	10

ii)	Create a Swift program that manages a guest list for an event: 1. Store guest names in an Array (for ordered access) and a Set (to avoid duplicates). 2. Add, remove, and display guests. 3. Show how both data structures behave when the same name is added twice.	10
Q.3	Attempt the following	20
i)	Design an Feedback Form screen using **UIKit** and **Storyboard** that includes the following UI components and layout specifications: – A `UILabel` titled **"Feedback Form"** at the top, cantered and bold. – A `UITextField` for **Name** with a placeholder. – A `UITextView` for **Comments** with a border and placeholder. – A `UILabel` showing **"Rate our service (1–5):"** – A `UISlider` (min = 1, max = 5, step = 1) to select a rating. – A `UILabel` that dynamically displays the selected rating value from the slider. – A `UIButton` labeled **Submit** at the bottom. • All elements must be organized inside a **vertical** `UIStackView`. • Use **Auto Layout** constraints to make the form responsive on iPhones and iPads. • When Submit is tapped , **navigate to a second screen** using a **push segue**. Pass the **name**, **rating**, and **comment** to the second screen using `prepare(for:sender:)`.	